

Technical Code Documentation

Dataset Table Name : Hospital

Power Query – Calculated Columns

Length of Stay Column

Calculates number of days between admission and discharge.

Length of Stay =

Duration.Days([Discharge Date] - [Date of Admission])

Age Group Segmentation

Categorizes patients into demographic groups.

Age Group =

if [Age] < 18 then "Child"

else if [Age] <= 35 then "Young Adult"

else if [Age] <= 55 then "Adult"

else "Senior"

Billing Category Classification

Segments billing amounts into cost categories.

Billing Category =

if [Billing Amount] < 15000 then "Low"

else if [Billing Amount] <= 30000 then "Medium"

else "High"

DAX Measures – Core KPIs

Total Patients

Counts total patient admissions.

Total Patients =

COUNTROWS(Hospital)

Total Revenue

Calculates overall billing revenue.

Total Revenue =

SUM(Hospital[Billing Amount])

Average Billing Amount

Calculates average billing per patient.

Average Billing =

AVERAGE(Hospital[Billing Amount])

Average Length of Stay

Measures average hospital stay duration.

Average Length of Stay =

```
AVERAGE(Hospital[Length of Stay])
```

Emergency Admissions

Counts emergency admission cases.

Emergency Admissions =

```
CALCULATE(  
    COUNTROWS(Hospital),  
    Hospital[Admission Type] = "Emergency"  
)
```

Patient Demographic Measures

Male Patients

Male Patients =

```
CALCULATE(  
    COUNTROWS(Hospital),  
    Hospital[Gender] = "Male"  
)
```

Female Patients

Female Patients =

```
CALCULATE(  
    COUNTROWS(Hospital),  
    Hospital[Gender] = "Female"  
)
```

Clinical Analysis Measures

Abnormal Test Results

Tracks abnormal diagnostic results.

Abnormal Tests =

```
CALCULATE(  
    COUNTROWS(Hospital),  
    Hospital[Test Results] = "Abnormal"  
)
```

Financial Analysis Measures

Revenue Per Patient

Measures profitability per admission.

Revenue per Patient =

```
DIVIDE(  
    [Total Revenue],  
    [Total Patients]  
)
```

High Billing Patients

Counts patients with high billing category.

High Billing Patients =

```
CALCULATE(  
    COUNTROWS(Hospital),  
    Hospital[Billing Category] = "High"  
)
```

Insurance Provider Count

Counts unique insurance providers.

Insurance Count =

```
DISTINCTCOUNT(Hospital[Insurance Provider])
```

Advanced Analytical Measures

Patients by Age Group

Patients by Age Group =

```
COUNTROWS(Hospital)
```

(Used with Age Group as axis)

Revenue by Hospital

Revenue by Hospital =

```
SUM(Hospital[Billing Amount])
```

(Used with Hospital column as category)

Patients by Medical Condition

Patients by Condition =

```
COUNTROWS(Hospital)
```

(Used with Medical Condition axis)

Medication Usage Count

Medication Count =

```
COUNTROWS(Hospital)
```

(Used with Medication as axis)

Tabular Summary Measures (For Tables & Matrix)

Patients per Insurance Provider

Patients by Insurance =
COUNTROWS(Hospital)

Revenue by Insurance Provider

Revenue by Insurance =
SUM(Hospital[Billing Amount])

Time-Based Analysis Measure

Patient Admission Trend

Patient Trend =
COUNTROWS(Hospital)
(Used with Date of Admission in line charts)

Supporting Calculated Logic Explanation

Length of Stay

Used to analyze:

- Treatment efficiency
- Resource utilization
- Patient recovery patterns

Age Group

Used for:

- Demographic analysis
- Healthcare demand segmentation

Billing Category

Used for:

- Cost distribution analysis
- High-value patient identification

Data Transformation Notes

The following data cleaning techniques were applied:

- Text trimming and formatting
- Date format standardization
- Removal of duplicate naming inconsistencies
- Creation of segmentation columns for enhanced analytics

Project Technical Summary

This project applies:

- Power Query transformations for data preparation
- DAX calculations for KPI development
- Aggregation logic for business insights
- Data segmentation for demographic and financial analysis
- Time intelligence for admission trend evaluation

===== End of Technical Code Documentation =====